THE UNCONDITIONAL TRANSLATE-BASIS PROBLEM HAS A NEGATIVE ANSWER

LITERATURE-STATUS PACKET FOR ARXIV:0906.1162

1. The source problem

Odell–Sari–Schlumprecht–Zheng ask whether, for $4 < p < \infty$, there are $f \in L_p(\mathbb{R})$ and $\Lambda \subseteq \mathbb{R}$ such that the translates $(T_\lambda f)_{\lambda\in\Lambda}$ form an unconditional basis of all $L_p(\mathbb{R})$. In the official arXiv PDF this is Problem 4.6 on page 27:

> **Problem 4.6.** Let $4 < p < \infty$. Does there exist $f \in L_p(\mathbb{R})$ and $\Lambda \subseteq \mathbb{R}$ so that $(f_{(\lambda)})_{\lambda\in\Lambda}$ is an unconditional basis for $L_p(\mathbb{R})$?

The question is nontrivial in precisely this range: the source constructs, for $p > 4$, unconditional basic sequences of translates whose closed span contains an isomorphic copy of $L_p(\mathbb{R})$, but that does not make the span equal to the ambient space or complemented in it.

2. The later answer

Freeman–Odell–Schlumprecht–Zsák prove a stronger negative theorem in arXiv:1209.4619. Their Corollary 2.3 (official PDF page 5) states:

> **Corollary 2.3.** *If $f \in L_p(\mathbb{R})$, $1 \le p < \infty$, $(T_\lambda f)_{\lambda\in\Lambda}$ is an unconditional basis for $X_p(f, \lambda)$ then $X_p(f, \Lambda) \ne L_p(\mathbb{R})$.*

Thus, for every $1 \le p < \infty$, if the translates of one f are an unconditional basis of their closed span $X_p(f, \Lambda)$, then

$$X_p(f, \Lambda) \ne L_p(\mathbb{R}).$$

Taking $4 < p < \infty$ gives an immediate negative answer to the source problem. The later paper explicitly describes its contribution as "filling the gap $(4, \infty)$": earlier results already ruled out an unconditional translate basis for $1 \le p \le 4$.

3. Identification and scope

The match is exact: both statements concern a single generator f, an arbitrary real translation set Λ, and an unconditional Schauder basis for the whole space $L_p(\mathbb{R})$. No additional separation, regularity, or ordering hypothesis is introduced in the answer.

This does *not* settle the source's main conditional Schauder-basis problem. In contrast, arXiv: 1209.4619 also constructs unconditional Schauder frames of translates for $p > 2$; redundancy is therefore the sharp distinction in that result.

References

[1] E. Odell, B. Sari, Th. Schlumprecht, and B. Zheng, *Systems formed by translates of one element in $L_p(\mathbb{R})$*, arXiv:0906.1162.

[2] D. Freeman, E. Odell, Th. Schlumprecht, and A. Zsák, *Unconditional structures of translates for $L_p(\mathbb{R}^d)$*, arXiv:1209.4619, Corollary 2.3.

Date: 11 August 2026.